Expo and Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected final evaluation and defence record.

Archive the candidate, collect both methods' actual results, prepare the defence and judge the independent library comparison.

Keep one habit: Keep saved versions, actual traces, full-pass counts, unrun cases and next actions together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Improve One Cause

Test Normal Cases

Choose Inputs and Outputs

W1 · Archive the fixed candidate before final inputs

> Use the supplied contract and record known limitations honestly.

Your evidence record

- Final-evaluation contract: whole available minutes 0–30 and Quiet required Yes/No; No permits either quiet label. A fit requires duration≤minutes AND (quiet not required OR activity quiet). Search in assigned catalogue order, stop with the first fit and reason, report no match only after every row fails, clarify invalid/missing fields before search and reset each new run. The teacher supplies a fresh labelled catalogue and requests only after the candidate instructions are saved. Both candidate and manual baseline use those same inputs and the same full-pass rule: correct output and supported reason. Keep actual observations separate from the answer key and leave unrun cases unscored.

- Before final cases are revealed, record your version label, where its complete instructions are saved, the agreed **input** and matching rules, the manual **baseline** and any known **limitation**.

  Confirm that the instructions stay unchanged while the assigned catalogue is used as **input**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Keep both methods' actual final records

> Record outputs, reasons, versions and failures before checking the key.

Your evidence record

- Use a separate sheet for the teacher's reserved cases. For every case record its ID and **input**s, catalogue and candidate labels, both methods' observed **output**s and reasons, and results after checking. Preserve failure traces.

  Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Prepare an evidence-based defence

> Use one actual trace, measured comparison, limitation and next action.

Your evidence record

- Write a two-minute explanation of the problem, one trace, your actual comparison, a **limitation** and a next action. Explain why this is a written-rule design rather than a model trained on examples.

  State which personal information is unnecessary and who decides whether to act.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Judge a different project independently

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Judge a different project independently

MAIN LESSON · SECTION 16

W5 · Write the final bounded conclusion

> Use your own records and retain privacy and human-control choices.

Your evidence record

- Use your own records to complete: Under these conditions, our saved version showed… The **baseline** showed… We still do not know… Our next action is… Include one privacy or human-control choice.

  If a case failed, explain its consequence without deleting the result.

My reasoning and evidence

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Present the final Classroom Sorter portfolio and complete the AI-1 certification **defense**.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Keep the **final evaluation** record for Evaluating and explaining your project. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an **evidence**-based statement and name a next test.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 36 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the **evidence**?

> **My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional library adviser and its manual **baseline** face the same 6 reserved cases. The candidate has 4 full passes; the **baseline** has 5. One additional planned case is unrun by both. No speed data was collected.

The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

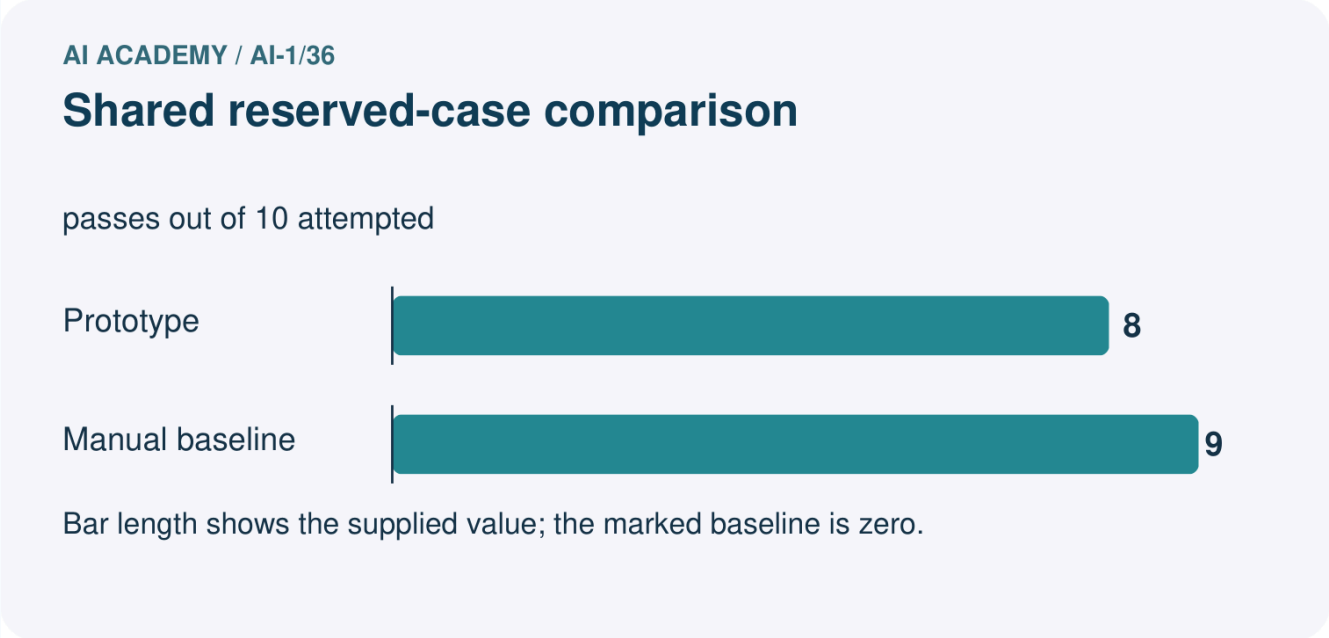
> Keep the explanation and the support you used.

Use this when helpful

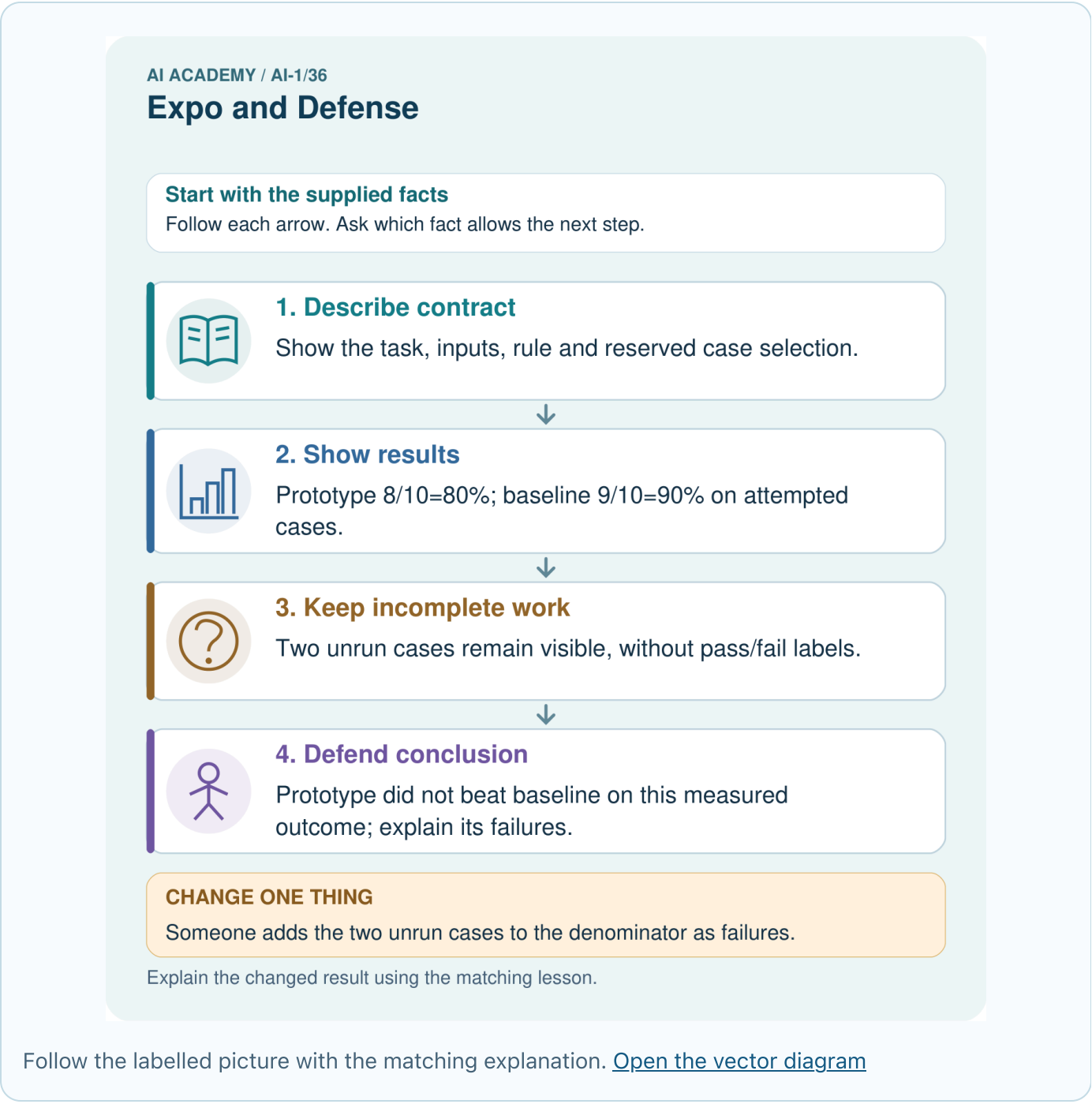
A fictional paper sorter team prepares an exhibition **evidence** board.

Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Write one defensible success statement and one claim the **evidence** cannot support.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional paper sorter team prepares an exhibition **evidence** board.
- Which labelled step matches this detail? Prototype did not beat **baseline** on this measured outcome; explain its failures.
- 1. Keep incomplete work
- 2. Describe contract
- 3. Show results
- 4. Defend conclusion
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Defend conclusion Prototype did not beat **baseline** on this measured outcome; explain its failures. An exhibition should make the **evidence** inspectable.

A polished **demonstration** is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster.

Unrun cases must not be treated as successes.

Changed-case question

- Changed-case practice: Someone adds the two unrun cases to the denominator as failures. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. That reports a different measure, 8/12 completed successes out of planned cases.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That reports a different measure, 8/12 completed successes out of planned cases.

It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Repair and save your work

Review Improve One Cause (ai-1/35). Goal: Repair a located defect without changing the agreed requirements. Short practice: Explain why “our repaired version now picks something” is not enough to show success at 6..

Return to this week and explain what you changed.

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards..

Return to this week and explain what you changed.

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

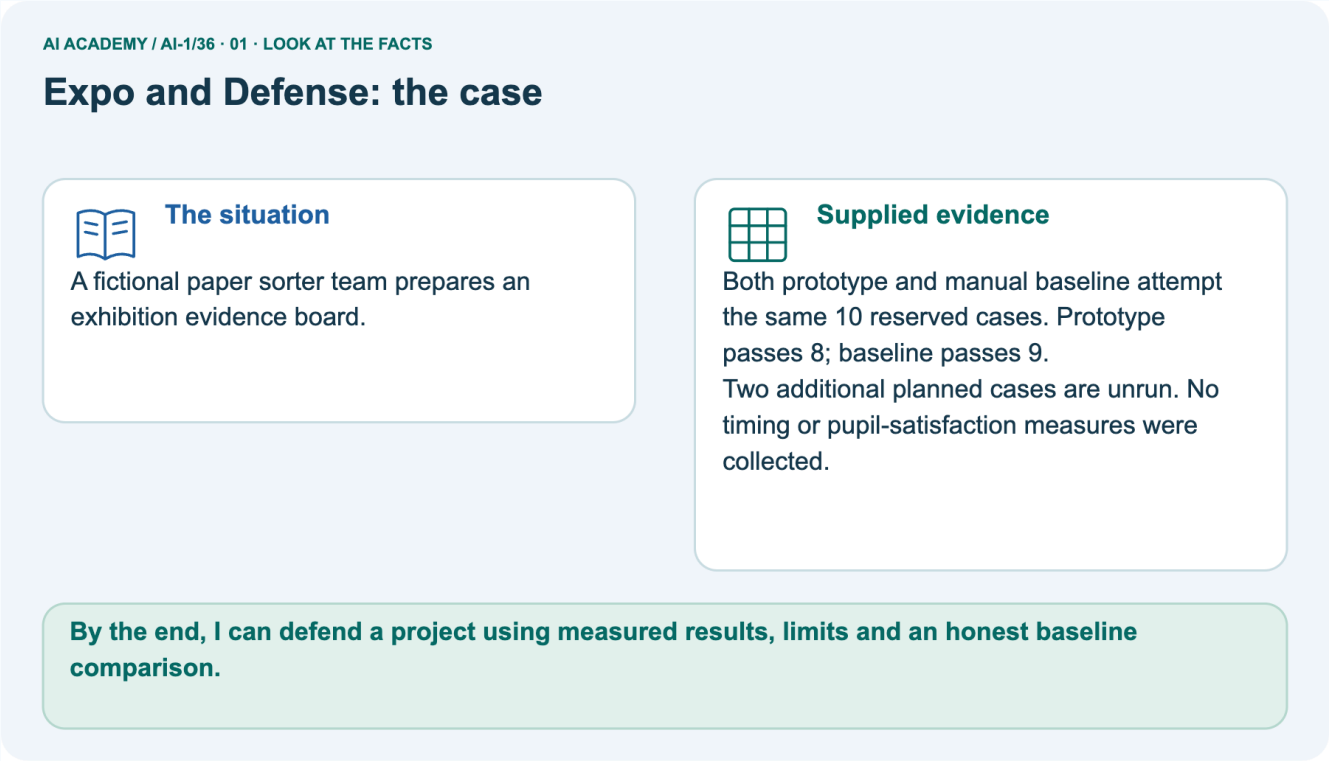
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Expo and Defense: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper sorter team prepares an exhibition **evidence** board. Supplied facts: Both prototype and manual **baseline** attempt the same 10 reserved cases. Prototype passes 8; **baseline** passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



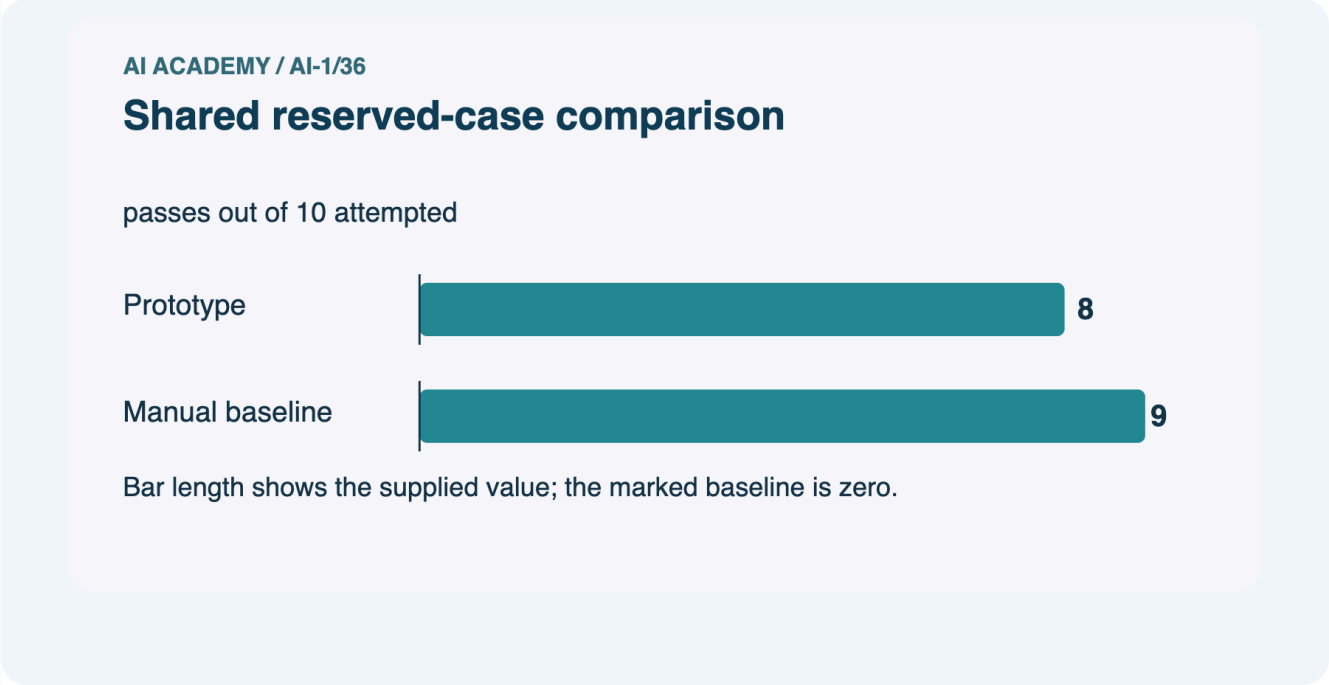
Follow the labelled picture with the matching explanation. Open the vector diagram

Describe contract: Show the task, **input**s, rule and reserved case selection. → Show results: Prototype 8/10=**80%**; **baseline** 9/10=90% on attempted cases. → Keep incomplete work: Two unrun cases remain visible, without pass/fail labels. → Defend conclusion: Prototype did not beat **baseline** on this measured outcome; explain its failures.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Shared reserved-case comparison



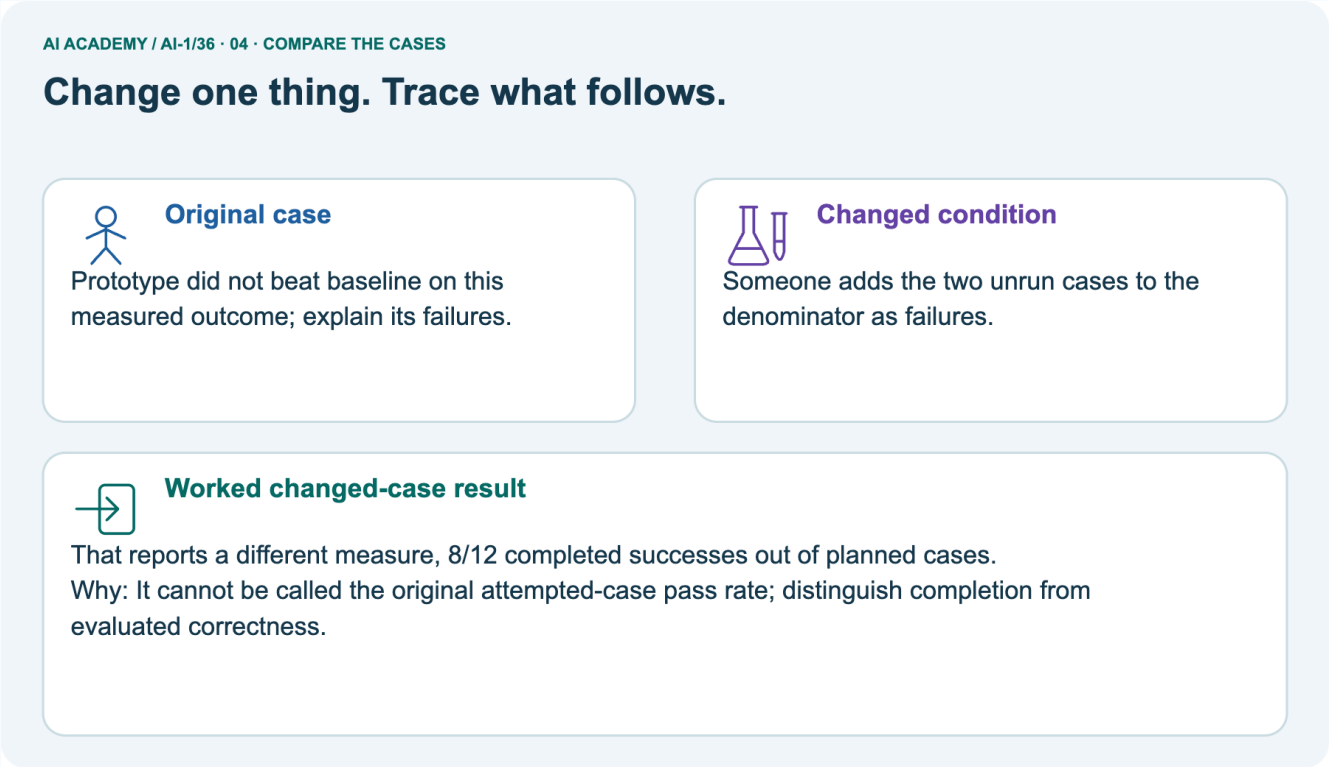
Follow the labelled picture with the matching explanation. Open the vector diagram

Shared reserved-case comparison. Units: passes out of 10 attempted. Prototype: 8. Manual **baseline**: 9.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

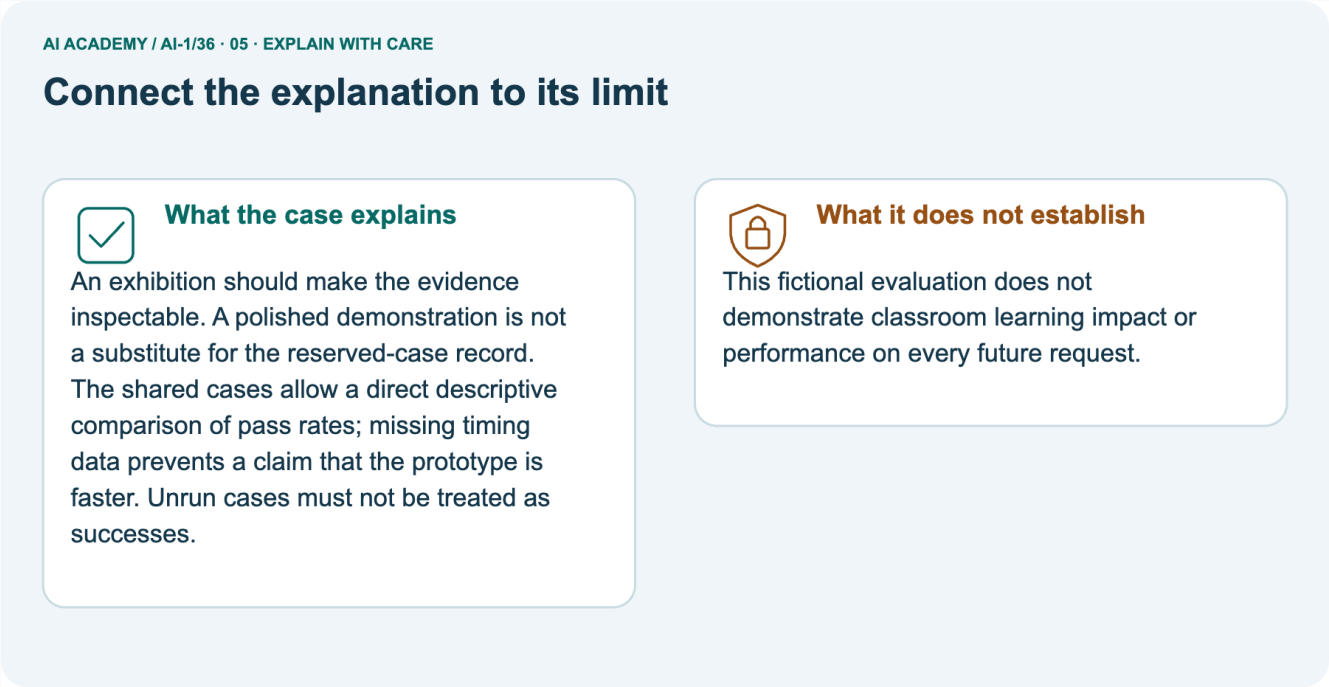
Original case: Prototype did not beat **baseline** on this measured outcome; explain its failures.
Changed condition: Someone adds the two unrun cases to the denominator as failures.

Expected result: That reports a different measure, 8/12 completed successes out of planned cases. Reason: It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An exhibition should make the **evidence** inspectable. A polished **demonstration** is not a substitute for the reserved-case record.

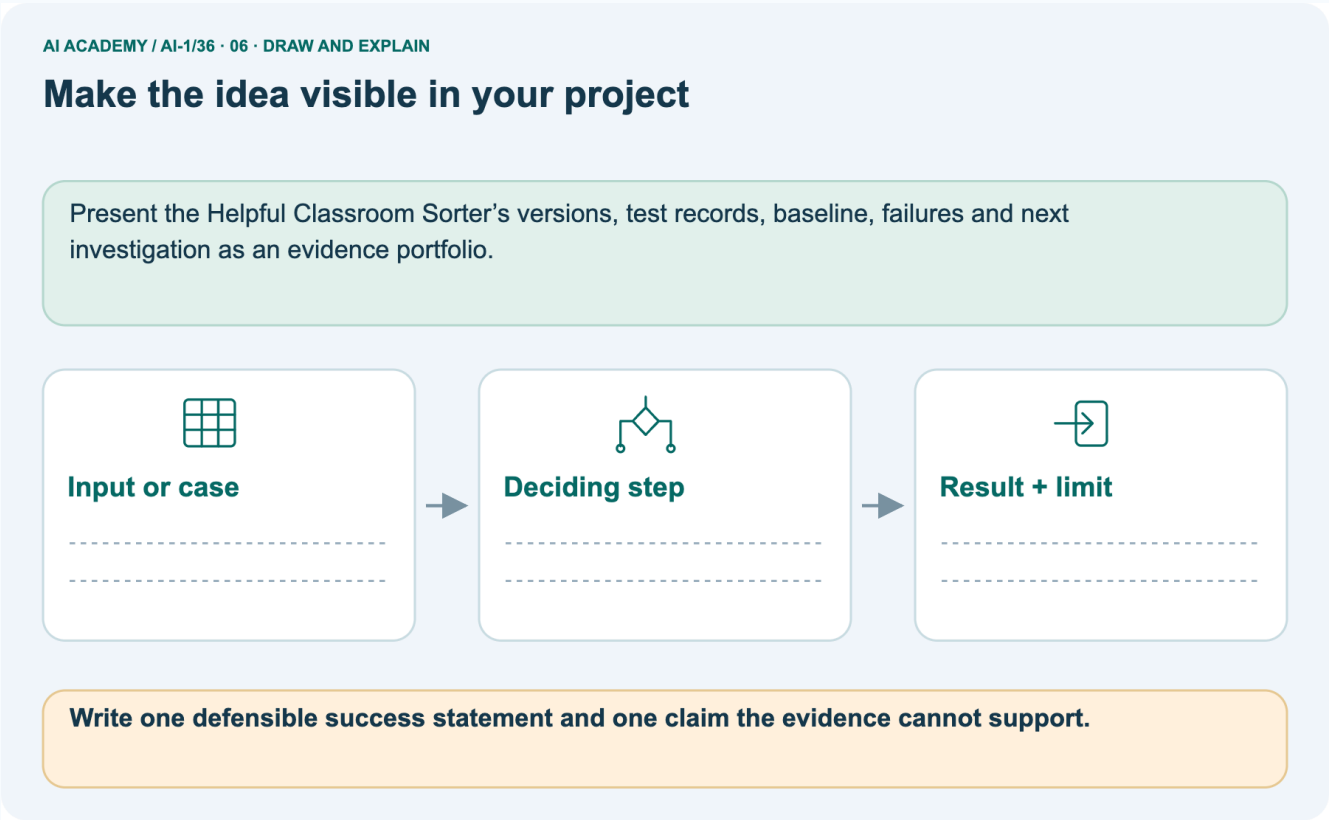
The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Limit: This fictional evaluation does not demonstrate classroom learning impact or performance on every future request.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Helpful Classroom Sorter's versions, test records, **baseline**, failures and next investigation as an **evidence** portfolio. Draw your **input**, deciding step, result and limit.

Supported question: Write one defensible success statement and one claim the **evidence** cannot support.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Expo and Defense: the case

02. Follow the deciding steps

03. Shared reserved-case comparison

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.